

Load Interconnection Request System Impact Study Report

For Queue Project Q14-0308

Pittsburg 138 kV
September 21, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0308. The Interconnection Customer proposes a 29.8 MW large load (data center) facility interconnecting at the Pittsburg 138 kV substation in the Pennsylvania area. The request was received on February 12, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 178 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,338,000.

3. Network Impacts

3.1 Load Deliverability Analysis

JOHNS CREEK - PRIMM 138 KV: 129.8% loading (188 MVA) following the outage of the LIONS ROAD - PINEYCRK PROJECT 500 KV; DFAX = 11.6%.

The KNOTT - PRIMM 138 KV is overloaded to 107.0% of its 184 MVA emergency rating for the loss of the LIONS ROAD - PINEY CREEK PROJECT 500 KV. The project contributes 14.7% DFAX (4.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
JOHNS CREEK - PRIMM 138 KV	Rebuild line with higher-capacity conductor	\$8,288,000	14% (\$1,187,000)
KNOTT - PRIMM 138 KV	Reconductor line	\$1,126,000	13% (\$151,000)
	Total allocated to project		\$1,338,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.